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(54) **MODULAR REVERSIBLE WATER HEATER DESIGN**

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F24H 9/40 (2022.01)

F24H 9/45 (2022.01)

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See application file for complete search history.

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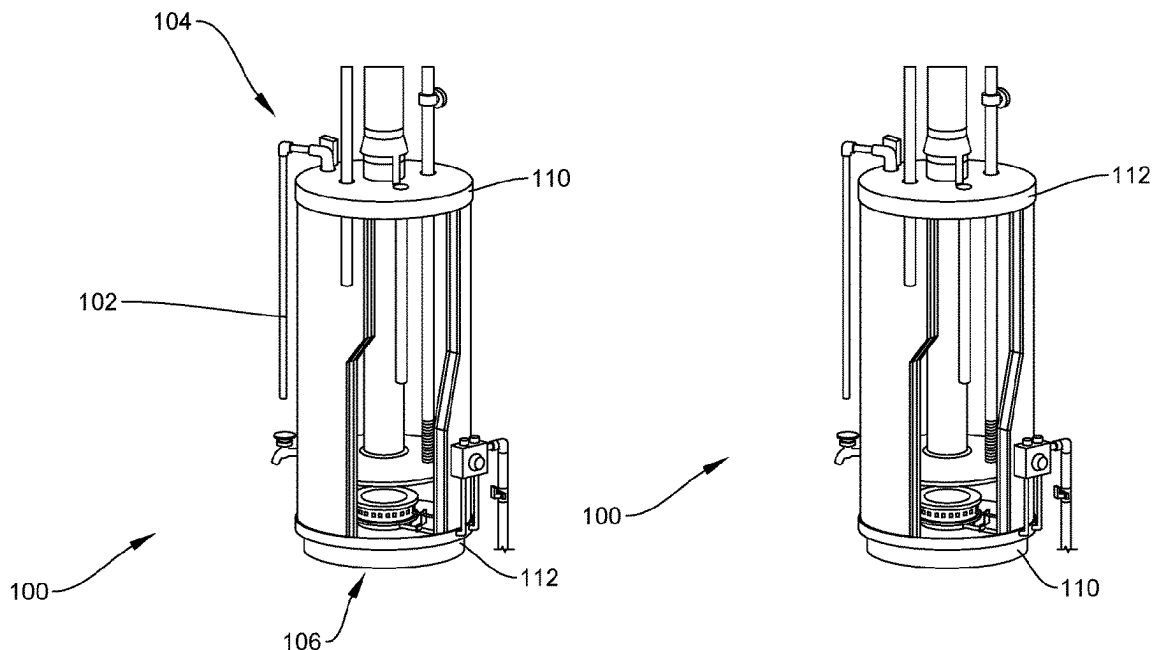
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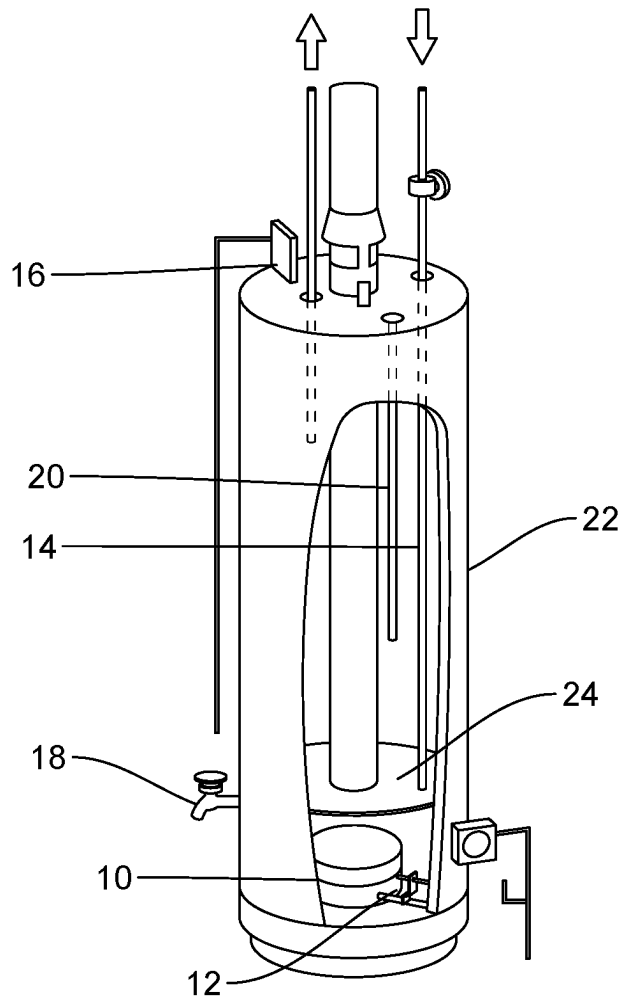
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ABSTRACT

A modular water heater tank design is provided that enhances the longevity and efficiency of domestic hot water heaters. Interchangeable upper and lower pans allow the water tank to be inverted after several years of continuous use. By repositioning the pans, the thermal stress and temperature gradients experienced by the tank are significantly reduced, thereby extending its operational life. Additionally, the salt precipitation within the tank is removed from new lower pan, ensuring improved heat absorption and preventing premature corrosion.

4 Claims, 2 Drawing Sheets





PRIOR ART
FIG. 1

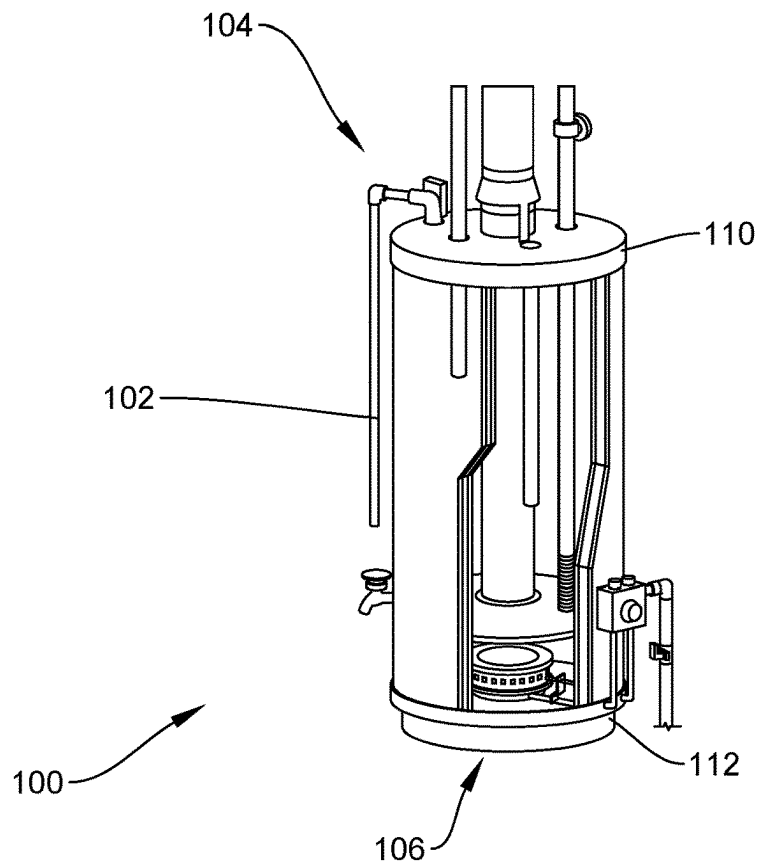


FIG. 2

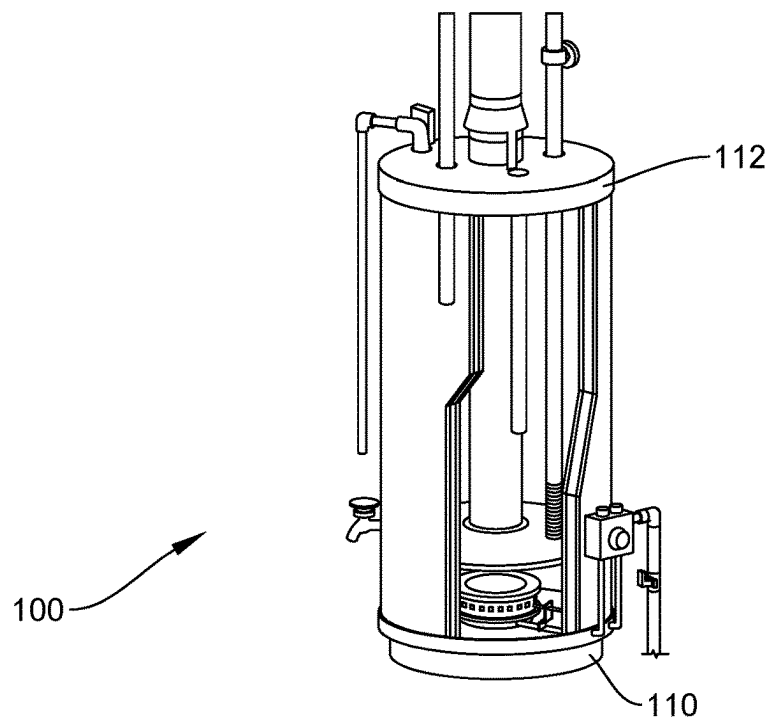


FIG. 3

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MODULAR REVERSIBLE WATER HEATER DESIGN

RELATED APPLICATIONS

There are no previously filed, nor currently any co-
pending applications, anywhere in the world.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates generally to a hot water
heaters for domestic use and, more particularly, to a modu-
larly constructed hot water heater that allows for increased
effective life.

2. Description of the Related Art

A hot water heater, also known as a water heater or geyser,
is a common appliance used to heat and store water for
various purposes such as domestic use, bathing, and space
heating. A domestic hot water heater consists of a relatively
large tank made of steel, usually about 1/8 of an inch thick.
Attached to the bottom holds a mechanism for generating
heat, whether by electric heat or a gas flame. There are some
chimneys running through the tank that act as heat
exchanger between the hot gasses and the water in the tank.
There is a second pan on the top of the tank that conducts the
products of combustion away from the tank and into pipes
to carry the fumes to discharge outside of the dwelling.

While different types of hot water heaters exist, as shown
in conjunction with FIG. 1 the following are typical com-
ponents and construction of a typical residential storage tank
water heater according to the PRIOR ART.

Heating Element/Burner. The heating element or burner
10 is responsible for heating the water in the tank 22. In an
electric water heater, one or two electric heating elements,
similar to large coils, are immersed in the water. These
elements heat up when electricity passes through them. In a
gas water heater, a gas burner is located at the bottom of the
tank, which ignites and heats the water when gas is supplied.

Thermostat. The thermostat 12 is a temperature control
device that regulates the water temperature in the tank. It is
usually located on the outside of the tank and allows the user
to set the desired temperature. When the temperature drops
below the set point, the thermostat signals the heating
element or burner to activate and heat the water until the
desired temperature is reached.

Dip Tube. The dip tube 14 is a plastic or copper tube that
delivers cold water from the incoming water supply to the
bottom of the tank. This ensures that cold water is introduced
at the bottom, while hot water remains at the top, ready for
use. It prevents the incoming cold water from mixing
directly with the hot water near the outlet.

Pressure Relief Valve. The pressure relief valve 16 is a
safety device designed to prevent excessive pressure buildup
inside the tank. It is typically located on the side or top of
the tank. If the pressure or temperature inside the tank
exceeds a certain limit, the valve automatically opens to
release the excess pressure and protect the tank from poten-
tial damage.

Drain Valve. The drain valve 18 is located at the bottom
of the tank and allows for easy draining of the water heater
when maintenance or repairs are required. It is typically a
simple faucet-like valve that can be opened to drain the
water from the tank.

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Anode Rod. The anode rod, also known as a sacrificial
anode 20, is a metal rod typically made of aluminum or
magnesium that is suspended inside the tank. Its purpose is
to attract and corrode more readily than the steel tank, thus
protecting the tank from rust and corrosion. Over time, the
anode rod may need to be replaced to ensure the longevity
of the water heater.

Tank. The tank 22 is a cylindrical or rectangular container
made of steel or other durable materials. It is designed to
store and hold a certain amount of water, typically ranging
from 20 to 80 gallons (75 to 300 liters). The tank 22 is also
generally insulated to minimize heat loss and maintain the
temperature of the stored water. The tank is enclosed to
provide a pressure-resistant water storage tank. In the center
of the tank is a hollow exhaust flue through which exhaust
gases from the burner flow up to an exhaust vent. In most
designs, a spiral metal baffle 24 inside the flue captures heat
from the exhaust gases and transmits it to the surrounding
tank.

The tank is often constructed using steel with a glass
lining or a glass-lined steel tank. This combination provides
durability and corrosion resistance. The steel tank provides
strength and structural integrity, while the glass lining helps
protect the steel from the corrosive effects of water and
extends the lifespan of the tank. The glass lining is usually
applied to the interior surface of the steel tank through a
vitreous enamel process, creating a smooth and protective
barrier between the water and the steel. This construction
ensures that the tank can withstand the pressure and tem-
perature of the hot water while resisting rust and corrosion.

In natural gas systems, a gas flame is utilized having a
temperature of approximately 2250° F., and the water com-
ing into the tank ranges near ambient, or approximately 50°
F. on average. Even at equilibrium, the water in the tank is
kept below boiling so a tremendous thermal gradient exists.
The steel in the base of the tank and the bottom of the
chimneys is significantly expanded by the heat of the flame
playing directly on the steel at the same time, on the other
side of the same steel plate, the steel plate and chimneys are
chilled and contracted by the water in the tank.

Throughout the appliance's functional life, the same sheet
of steel is expanded and contracted at the same time by the
extreme temperature gradient. The steel is effectively wrung
by trying to go in opposite directions at the same time. The
result is extreme stress on the steel. In a few years the steel
splits and the water heater is lost, as well as some peripheral
damage to floors and furniture in the home.

Another problem is that sparingly soluble salts within the
water will precipitate out at the bottom of the tank as a solid
layer. At first this layer reduces heat absorption and valuable
heated air drifts out the chimney. Later it becomes so thick
that it cuts off absorption completely. At this point the steel
is heated to the temperature of the flame and is constantly
heated to 2250° F. and then cooled, over and over with each
cycle. Such temperatures are close to the melting point of
steel, and steel does not last long in such extreme conditions.
And, when the tank fails there is a flood in the home, so the
cost of a replacement water heater also is generally accom-
panied by the cost of cleanup and other damage.

Some methods and devices are known that incorporate
various mechanisms for minimizing damage caused by
thermal cycling and salt precipitate. For example:

U.S. Pat. No. 2,718,583 issued in the name of Nolan et al.
describes a water-heater tank of reinforced plastic and
method and apparatus for making the same. The plastic
water heater tank is formed by helically winding subse-
quently setting-plastic-impregnated glass-fiber roving cords

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over and around convex tank heads and helically around a cylindrical lining shell connected there between. An electric resistance wire may also be embedded in the tank walls.

U.S. Pat. No. 4,865,014, issued in the name of Nelson, describes a water heater which may be either gas or electric and designed for use in either residential or commercial applications includes a generally cylindrical molded plastic inner water tank and a generally cylindrical molded plastic outer shell which is sized somewhat larger than the outside diameter of the tank and is disposed around the tank so as to define an annular clearance space between the shell and the tank. By means of covers and end members the clearance space between the tank and the shell are sealed closed so that a vacuum can be drawn on this closed clearance space and thereby provide thermal insulation for the inner water tank by means of the vacuum. A related construction includes a double-walled outer shell wherein the two walls are spaced apart from one another so as to define a clearance space there between. This clearance space is sealed closed and a vacuum drawn in order to provide the requisite insulation for the inner water tank which is disposed on the interior of the shell.

And, Canadian Patent No. 2,759,310, issued in the name of Lesage et al., describes a method of constructing an inner glass-lined steel tank. The lower edge of the top dome shell is machined wherein a smooth angulated surface is formed between an outer and an inner surface of the top dome shell to form a smooth angulated surface to eliminate abrupt transition edges. Also, couplings are machined and press-fitted into the tank wall to form smooth surfaces with the inner surface of the tank. When the porcelain enamel is sprayed inside the tank there are no sharp edges which often results in defects due to the fact that the porcelain enamel is not properly adhered thereto. Such defects eventually cause corrosion within the inner tank and greatly reduce the life expectancy thereof as well as the life expectancy of the sacrificial anode connected to the tank and extending therein.

Consequently, a need has been felt for hot water heater design that allows for a means to allow for extended operational life of the device.

SUMMARY OF THE INVENTION

It is thus an object of the present invention to provide an improved hot water heater designed for extended operational life.

It is a feature of the present invention to provide a modular reversible water heater design that allows for changing of hot and cool side components prior to their failure.

The present invention provides a modular water tank design in which the upper and lower pans are made such that they are interchangeable. After several years of continuous use, the top and bottom pans may then be removed, and the water tank inverted, and the pans reattached on the opposite ends of where they were initially installed. Once inverted, the original bottom steel plate is now positioned well away from the hot flame to stop additional thermal cycling. Further, the salt-covered bottom of the hot water heater is now on the top, where the temperature gradient is lesser. The new lower tank pan now has a history of less thermal cycling, and the newly configured tank will now sustain an additional several years of operation before succumbing to failure.

An advantage of the present invention includes extended lifespan, with the modular reversible design significantly

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extending the operational life of water heaters by reducing thermal stress and addressing salt accumulation.

Further, the present invention minimizing extreme temperature gradients and enhances the efficiency of the water heater, leading to improved heat absorption and reduced energy consumption.

Further still, the invention reduces maintenance costs and the need for premature replacements, resulting in substantial savings for homeowners. And, the longer-lasting water heaters contribute to reduced waste and environmental impact associated with frequent replacements.

Further objects, features, elements and advantages of the invention will become apparent in the course of the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

The advantages and features of the present invention will become better understood with reference to the following more detailed description and claims taken in conjunction with the accompanying drawings, in which like elements are identified with like symbols, and in which:

FIG. 1 is a schematic diagram of an otherwise conventional gate water heater design according to the PRIOR ART;

FIG. 2 is a schematic diagram of an improved modular hot water heater design according to the preferred embodiment shown in a first orientation; and

FIG. 3 is a schematic diagram thereof shown in a second orientation.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The best mode for carrying out the invention is presented in terms of its preferred embodiment, herein depicted within the Figures. It should be understood that the legal scope of the description is defined by the words of the claims set forth at the end of this patent and that the detailed description is to be construed as exemplary only and does not describe every possible embodiment since describing every possible embodiment would be impractical, if not impossible. Numerous alternative embodiments could be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims. Finally, unless a claim element is defined by reciting the word "means" and a function without the recital of any structure, it is not intended that the scope of any claim element be interpreted based on the application of 35 U.S.C. § 112 (f).

The best mode for carrying out the invention is presented in terms of its preferred embodiment, herein depicted within the FIG. 2 through FIG. 3.

1. Detailed Description of the Figures

Referring now to the drawings, wherein like reference numerals indicate the same parts throughout the several views, an improved modular reversible water heater tank, generally noted as **100**, is provided according the preferred embodiment of the present invention. The tank **100** comprises a cylindrical or rectangular sidewall **102** for containing a working volume of water. The sidewall has an upper end **104** opposite a lower end **106**.

The upper end **104** is equipped with an upper pan **110**. The lower end **106** is equipped with a lower pan **112**. The pans

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110, 112 are specifically adapted to be removed and reattached on the opposite ends of the tank 100 after a period of continuous use.

The tank sidewall 102 may consist of a steel outer jacket that encloses a pressure-tested water storage tank. This inner tank may be made of high-quality steel with a vitreous glass or plastic layer bonded to the inside surface to prevent rusting. An exhaust flue is directed through the upper pan 110 for discharge of exhaust gases to an exhaust vent. In most designs, the tank 102 may also generally be formed of an inner storage tank and the outer tank jacket with a layer of insulation therebetween designed to reduce heat loss (not shown). Both the upper pan 110 and the lower pan 112 may also be made of steel or of steel with a glass or plastic lining or a glass-lined steel tank.

The pans 110, 112 are further affixed to and seal the tank sidewall 102 such as to ensure that the tank can withstand the pressure and temperature of the hot water while resisting rust and corrosion.

2. Operation of the Preferred Embodiment

In operation, a reconfiguration of the modular tank design effectively changes the orientation of the tank, minimizing the impact of extreme temperature gradients and thermal stress. As shown in conjunction with FIG. 3, after a number of years in operation, the placement of the pans 110, 112 are reversed in their location on the tank, ensuring that the original lower pan 112, which is susceptible to extreme thermal cycling, is relocated to the upper end 104 and away from the direct heat source. Similarly the original upper pan 110, having been exposed to reduced stress on the material, is positioned at the lower end 106 and thereby extending the tank's operational life.

By repositioning the upper and lower pans, the invention mitigates the simultaneous expansion and contraction of the steel caused by extreme temperature gradients. This reduction in thermal stress significantly prolongs the life of the tank. Further, the modular design also addresses the issue of salt accumulation within the tank. By inverting the tank and enclosing pans, the salt-covered bottom is moved to the top, where temperature gradients are less severe. This prevents salt from interfering with heat absorption and reduces the risk of corrosion.

The foregoing descriptions of specific embodiments of the present invention are presented for purposes of illustration and description. The Title, Background, Summary, Brief Description of the Drawings and Abstract of the disclosure are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the Detailed Description, it can be seen that the description provides illustrative examples and the various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

The claims are not intended to be limited to the aspects described herein, but is to be accorded the full scope consistent with the language claims and to encompass all

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legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirement of 35 U.S.C. § 101, 102, or 103, nor should they be interpreted in such a way. Any unintended embracement of such subject matter is hereby disclaimed. They are not intended to be exhaustive nor to limit the invention to precise forms disclosed and, obviously, many modifications and variations are possible in light of the above teaching. The embodiments are chosen and described in order to best explain principles of the invention and its practical application, to thereby enable others skilled in the art to best utilize the invention and its various embodiments with various modifications as are suited to the particular use contemplated. It is intended that a scope of the invention be defined broadly by the Drawings and Specification appended hereto and to their equivalents. Therefore, the scope of the invention is in no way to be limited only by any adverse inference under the rulings of *Warner-Jenkinson Company, v. Hilton Davis Chemical*, 520 US 17 (1997) or *Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co.*, 535 U.S. 722 (2002), or other similar caselaw or subsequent precedent should not be made if any future claims are added or amended subsequent to this Provisional Patent Application.

What is claimed is:

1. A modular reversible water heater system comprising: a cylindrical liquid filled tank with an inner surface and an outer surface, said liquid filled tank having a tank head on a top and a tank base on a bottom and being constructed from a metal material;

an upper cap removably attached over the tank head and a lower heating apparatus enclosure removably attached under the tank base, wherein said upper cap includes a chimney structure configured for conducting products of combustion away from said liquid filled tank and into exhaust pipes, and said lower heating apparatus enclosure accommodates a heat-generating mechanism;

wherein after a predetermined period of use in an original orientation, the liquid filled tank is designed to be physically invertible relative to said upper cap and said lower heating apparatus enclosure, and upon said physical inversion, the upper cap and lower heating apparatus enclosure are reattachable to said liquid filled tank in an inverted orientation relative to the original orientation while maintaining the upper cap over the tank base and the lower heating apparatus enclosure under the tank head, said physical inversion repositioning the tank base away from the heat-generating mechanism and the tank head closer to the heat-generating mechanism, thereby mitigating thermal cycling effects on the metal material and optimizing positioning of accumulated salts.

2. A method for enhancing the longevity of a hot water heater, comprising the steps of:

providing a tank with an inner surface and an outer surface, said tank having a tank head on a top and a tank base on a bottom and being constructed from a metal material;

attaching an upper cap removably attached over the tank head and a lower heating apparatus enclosure removably attached under the tank base, said upper cap configured for conducting products of combustion away from said tank and into exhaust pipes, and said lower heating apparatus enclosure accommodating a heat-generating mechanism;

utilizing said hot water heater tank for a predetermined period of time;

detaching said upper cap and said lower heating apparatus enclosure from said tank;

physically inverting the tank relative to an original orientation such that the tank base is repositioned away from the heat-generating mechanism and the tank head is repositioned closer to the heat-generating mechanism; and

reattaching said upper cap to the tank base and said lower heating apparatus enclosure to the tank head such that the upper cap remains over the tank and the lower heating apparatus enclosure remains under the tank after said physical inversion, thereby mitigating the effects of thermal cycling on the metal material and optimizing positioning of accumulated salts.

3. The method of claim 2, further comprising the step of utilizing said upper cap to conduct products of combustion away from said tank and into exhaust pipes.

4. The method of claim 2, further comprising the step of utilizing said lower heating apparatus enclosure to house the heat-generating mechanism, wherein the heat-generating mechanism comprises an electric heating element or a gas burner for generating heat.

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